

Compact Embeddings for Eye Blink Detection on Edge Hardware

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Abstract—Camera-based eye blink monitoring is often deployed on edge-class hardware, where the cost of the visual pipeline matters as much as its accuracy. We detect blink events on a low-dimensional representation: each frame is encoded once into a 16-dimensional embedding by a compact convolutional encoder, and a temporal head classifies a buffer of these embeddings, costing 0.37% of the per-frame compute. On a 57-subject subset of the mEBAL2 database (27,758 events, subject-disjoint five-fold cross-validation, three seeds), the proposed model reaches a precision–recall AUC (PR-AUC) of 0.9886 ± 0.0038 against 0.9906 ± 0.0037 for an image-based convolutional baseline with the same temporal head. The paired subject-clustered difference, -0.0040 (95% CI $[-0.0063, -0.0020]$), lies within a non-inferiority margin of $\delta = 0.02$ fixed in advance, at $5.7\times$ fewer parameters and $2.6\times$ fewer multiply–accumulate operations. On a Raspberry Pi 5 the representation stage costs 0.86 against 2.05 ms per frame, yet end to end the control is only 11% slower (12.51 against 11.27 ms median), since landmark detection dominates the pipeline.

■ **SPONTANEOUS EYE BLINKING** is a low-cost behavioral signal used for ocular surface stress and attention estimation [1], [2]. The practical question is not whether blinks can be detected, but at what cost they can be detected continuously on the device that holds the camera.

Two families of methods are widely used. Landmark-based methods reduce each frame to a geometric ratio such as the eye aspect ratio (EAR) [3], which is negligible in cost once landmarks are available but compresses the eye region to a single scalar.

Image-based convolutional methods classify the eye crop directly [2], [4], retaining appearance information that a scalar discards, but pay a convolutional cost over the full crop at every frame. Since the original EAR formulation is itself a learned classifier over a temporal window of EAR values [3], we report both EAR variants to avoid understating that family.

We take an intermediate position (Figure 1): each frame is encoded once into an embedding $z \in \mathbb{R}^{16}$, buffered over the last 19 frames, and a temporal convolutional head classifies the window on embeddings, so re-encoding is never required.

Under a common crop, subject split and temporal

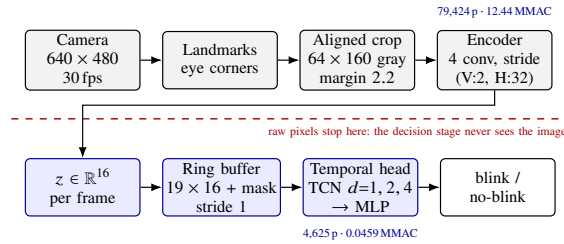


FIGURE 1. Proposed pipeline. Each frame is encoded once into a 16-dimensional embedding; the 19-frame window is assembled from buffered vectors rather than re-encoded, so the temporal head costs 0.37% of the per-frame compute.

head, we quantify the accuracy–cost trade-off between a compact encoder and an adapted image-CNN reference. Both learned frontends produce 16-dimensional features and both cache one representation per frame; the controlled comparison is therefore between backbones. Our contributions are the paired subject-level analysis of this trade-off, a transparent accounting of model and pipeline cost, and a Raspberry Pi 5 stage decomposition showing that landmark detection limits the end-to-end benefit. The stride and embedding-width ablations are exploratory checks, not evidence of a uniquely optimal architecture.

RELATED WORK

Blink detection has used landmark ratios and learned temporal windows [3], per-frame convolutional classifiers with score aggregation [2], sequence models such as recurrent and transformer architectures [4], [5], and dedicated in-the-wild benchmarks [6], [7], whose multi-person, untrimmed setting lies outside our single-user, event-level, edge-CPU scope. Embedded blink detection has also reached wearable hardware [8] and capacitive sensing [9], both of which differ from ours in sensing modality; closer to our approach, a 3-D autoencoder classifies blinks from a learned latent code [10], but for offline video analysis, aggregating per-frame predictions rather than one label per window, so it is not included as a baseline. We instead target a representation compact enough to measure its per-frame cost on an edge CPU.

METHOD

Overview

Figure 1 shows the pipeline: a rotation-aligned crop is encoded to $z \in \mathbb{R}^{16}$, the last 19 embeddings

are buffered, and a temporal convolutional head with a small multilayer perceptron (MLP) emits a blink score. Landmark coordinates are used only for the crop geometry and then discarded. The temporal head follows standard dilated 1-D convolutions. The same buffering strategy applies to the image-CNN reference; we measure the cost saved by replacing its backbone.

Eye Crop Normalization

Let $p_L, p_R \in \mathbb{R}^2$ be the landmark-derived eye centers (the midpoints of the eye corners), $d_{io} = \|p_R - p_L\|_2$, and $c = (p_L + p_R)/2$. We center a crop at c , rotate it so that the inter-ocular axis is horizontal, set its width to $w = 2.2d_{io}$, and resample it to a $H \times W = 64 \times 160$ grayscale image. Thus its source-image height is $h_{crop} = (H/W)w = 0.88d_{io}$; the scale is therefore normalized against camera distance. Pixel values are standardized using only the current frame,

$$\tilde{x}_t = \frac{x_t - \mu_t}{\max(\sigma_t, 0.255)}, \quad (1)$$

where μ_t and σ_t are the crop mean and standard deviation. The 0.255 floor equals $10^{-3} \times 255$ and prevents division by zero. Out-of-border regions are reflection-padded; resampling uses cubic interpolation when upsampling and area interpolation when downsampling.

Encoder Design and Vertical-Resolution Ablation

The encoder has four Conv–BatchNorm–ReLU blocks with channels (16, 32, 48, 64), kernels (3×5 , 3×3 , 3×3 , 3×3), and strides (1, 4), (2, 2), (1, 2), (1, 2) in (vertical, horizontal) order, with no bias term since each is followed by batch normalization. Same padding gives total strides of 2 vertically and 32 horizontally, hence a 32×5 map. Horizontal average pooling reduces this to a 64×32 feature, which a linear layer maps to $D = 16$. The encoder contains 79,424 parameters and costs 12.44 MMAC per frame, 12.41 of it convolutional.

A matched *v-drop* (reduced vertical resolution) variant keeps the kernels, channels and horizontal strides unchanged while raising the vertical stride from 2 to 16, cutting the convolutional cost from 12.41 to 3.44 MMAC; it is evaluated as an exploratory ablation under *Detection Accuracy*.

Temporal Modeling on Embeddings

Re-encoding all $T = 19$ frames at every decision would multiply the encoder cost by 19. Instead, each

frame is encoded once and the window is assembled from buffered vectors,

$$\mathbf{z}_t = f_\theta(\tilde{\mathbf{x}}_t) \in \mathbb{R}^D, \quad \mathbf{Z}_t = [\mathbf{z}_{t-T+1}^\top; \dots; \mathbf{z}_t^\top] \in \mathbb{R}^{T \times D}. \quad (2)$$

For every retained training/test event, $m_j \in \{0, 1\}$ records whether window position j is valid, with $\sum_j m_j \geq 14$. This is a dataset inclusion rule; a deployment system must implement its own explicit missing-frame policy. The head uses three residual 1-D Conv-BatchNorm-ReLU layers with kernel size 3 and dilations $d_\ell \in \{1, 2, 4\}$:

$$\begin{aligned} s_j^{(\ell)} &= \sum_{q=-1}^1 \mathbf{W}_q^{(\ell)} [m_{j-d_\ell q} \mathbf{h}_{j-d_\ell q}^{(\ell-1)}], \\ \mathbf{h}_j^{(\ell)} &= \text{ReLU}\left(\text{BN}_\ell\left(s_j^{(\ell)} + \mathbf{b}^{(\ell)}\right)\right) + \mathbf{h}_j^{(\ell-1)}, \end{aligned} \quad (3)$$

where $s_j^{(\ell)}$ is the masked convolutional sum, $\mathbf{h}_j^{(0)} = \mathbf{z}_j$, and out-of-window indices are zero-padded, the default of the 1-D convolution. The mask is applied to the buffer, before each convolution, and to the final layer output; masked positions are excluded from both pooling operations,

$$\mathbf{h}_{\text{avg}} = \frac{\sum_j m_j \mathbf{h}_j^{(3)}}{\max(1, \sum_j m_j)}, \quad \mathbf{h}_{\text{max}} = \max_{j:m_j=1} \mathbf{h}_j^{(3)}, \quad (4)$$

$$p = \varsigma(\text{MLP}([\mathbf{h}_{\text{avg}}; \mathbf{h}_{\text{max}}])), \quad (5)$$

where ς is the logistic function. The MLP has a 32-dimensional input, a 64-unit ReLU layer with dropout 0.3, and one logit output. Masking is not renormalized by the number of valid taps, and symmetric padding lets a position use later frames within its already collected window; the deployed decision requires no frames beyond the current buffer, though a centered window places its center nine frames (0.3 s at 30 fps) before the decision. This nominal alignment offset is not a measured onset-to-alert latency. The head costs 0.0459 MMAC per stride-1 decision, 0.37% of the 12.49 MMAC per-frame model total.

Reference Formulation

For the EAR-rule baseline, EAR_j is the mean EAR of both eyes at valid position j , and j_f, j_l the first and last valid positions in the window. We use the relative drop $r_t = 1 - \min_{j:m_j=1} \text{EAR}_j / [(\text{EAR}_{j_f} + \text{EAR}_{j_l})/2]$ and select its classification threshold on the validation fold.

EXPERIMENT

Dataset

We use the RGB webcam stream of a 58-subject subset of the mEBAL2 database [4], whose full release contains 180 subjects and near-infrared channels we do not use. The subset combines 38 subjects from the 2020 acquisition and 20 added in 2022. One subject (U18) is excluded (post hoc, not preregistered) because that session was recorded with several people in frame and, whenever the primary subject was not resolved, a neighboring person was cropped instead, affecting 31% of that subject’s events. The remaining 57 subjects yield 27,758 events (13,820 blink, 13,938 no-blink). Each event is a fixed 19-frame window, and events with more than five unresolved frames are excluded. The extraction stage stores 532,109 unique (subject, frame) crops — a pipeline total covering the excluded subject, later-discarded events, and crop reuse across overlapping events, not the product of the event and window counts.

Protocol

We use subject-disjoint five-fold cross-validation, stratified jointly by event count and acquisition batch, with three training seeds (15 runs). For rotation i , fold i is the test fold, fold $(i + 1) \bmod 5$ the validation fold, and the rest are used for training. A threshold is selected only on the validation fold, maximizing accuracy over all unique score values and their midpoints, then applied unchanged to the test fold. All methods share the folds, seeds, threshold-selection code, and bootstrap implementation, though the numeric grid is derived from each model’s own validation scores. The EAR rule is deterministic within a fold; EAR + head is a learned, seed-dependent model.

Training

All learned methods use binary cross-entropy with logits, Adam (learning rate 10^{-3}), batch size 32, up to 80 epochs with early stopping (patience 10 on validation PR-AUC), no data augmentation, and seeds 0, 1, 2 with deterministic settings enabled. The Image-CNN control is adapted from the mEBAL architecture [2] to our 64×160 grayscale two-eye crop and event labels (adaptation details under *Baselines*).

Baselines

Seven methods are compared. *EAR (rule)* thresholds the relative EAR drop within the window. *EAR*

+ *head* lifts four per-frame features (left/right EAR, their mean and minimum) to 16 channels via a $4 \rightarrow 16 \rightarrow 16$ BN+ReLU projection and feeds the same temporal head. *Image-CNN (max)* is the convolutional detector proposed by the database authors [2], with sequence scoring by the maximum of per-frame scores as in the original. *Image-CNN (+head)* replaces that aggregation with our temporal head, leaving only the frontend different. *Image-CNN (+BiLSTM)* and *Image-CNN (+LSTM)* keep the same Image-CNN backbone and swap in a single-layer recurrent head of hidden size 16 (matched to the temporal convolutional network (TCN) head) feeding the same MLP readout. BiLSTM runs forward and backward passes over the window. Both recurrent heads use masked mean and max pooling over all recurrent outputs, followed by an MLP; neither uses only the final hidden state. They see only the completed buffer, although BiLSTM uses both directions within it. The 16-D Image-CNN backbone has 471,536 parameters, and the added TCN, BiLSTM and LSTM heads have 4,625, 8,577 and 4,353 parameters, respectively. The max variant has a 470,561-parameter scalar-output backbone; its smaller projection must not be counted as a difference in temporal-head size. *Ours* uses the same 4,625-parameter TCN as Image-CNN (+head).

The recurrent accuracy rows were imported from a training handoff summary. Their raw per-run predictions and trained checkpoints were not available for this revision; we retain these rows as reported secondary results, not independently reanalysed evidence.

EAR is computed from the database's landmarks, so all methods share one detector and face-selection rule; we use the stronger relative-drop EAR variant (PR-AUC 0.17 higher than a minimum-EAR rule, which is not reported separately). The image-based backbone is not our design — it is the architecture the database authors proposed and benchmarked [2], reducing the concern that we chose a weak baseline — though it was originally trained with per-frame open/closed labels rather than our one-label-per-window protocol, so the max-pooling variant (trained under multiple-instance supervision) may underestimate the original method.

Metrics

The primary metric is PR-AUC, with blink as the positive class; ROC-AUC, accuracy, precision, recall, and F_1 are reported alongside. For each paired comparison we pool the held-out predictions of all 15

runs, compute one PR-AUC difference, and resample the 57 subjects with replacement 2,000 times (each draw carries that subject's three seeds), reporting percentile rather than bias-corrected 95% intervals. For a reference method, $\Delta = \text{PR-AUC}_{\text{ours}} - \text{PR-AUC}_{\text{ref}}$ is non-inferior when the lower limit of its paired 95% confidence interval exceeds $-\delta$, with $\delta = 0.02$ specified before the final comparisons.

Cost covers parameters, MMAC per frame, latency percentiles, peak memory, CPU utilization, idle-subtracted rail power and total-active rail energy per frame; operation counts include only convolutions and linear projections (excluding bias, batch norm, activations, pooling). One MAC is one multiply plus one add (two FLOPs); MMAC (10^6 MACs) is used throughout so encoder, head and edge-measurement costs are directly comparable.

Edge Platform

Measurements use a Raspberry Pi 5 (16 GB, active cooling), Debian 13, Python 3.11, and ONNX Runtime restricted to two intra-op threads with spinning disabled. The latency frontend uses MediaPipe FaceMesh with one face and `refine_landmarks=False`; the accuracy experiments use the meshes distributed with mEBAL2. With the CPU governor set to performance, six modes were replayed without frame-rate pacing, three times each. Although the runner requested 300 s, the 18 power runs ended at the video boundary after 98.65–130.39 s and 10,268 frames, achieving 78.75–104.09 fps. Thus these runs characterize saturated offline throughput, not power at a matched 30 fps camera rate. The configured PMIC sampling rate was 7.5 Hz; observed sampling was approximately 6.3 Hz after command overhead. The sum of 12 reported rails is provisional: the rail definition was not independently verified and is not a whole-board input-power measure. We report mean active-minus-idle power (30 s idle baseline), but energy per frame is the trapezoidal integral of total active rail power divided by processed frames. The two quantities must not be interpreted as the same idle-subtracted metric. No throttling flags were reported in these runs. Separate earlier 300 s latency runs provide the end-to-end medians quoted below; they are not the 18 power runs.

Detection Accuracy

Table 1 reports the runs. The best PR-AUC in the table is not ours: the image-based backbone with our

TABLE 1. Blink Detection Performance and Static Cost. Subject-disjoint 5-fold $\times$ 3 seeds; 57 subjects, 27,758 events. PR-AUC and ROC-AUC are mean $\pm$ SD over 15 runs; other metrics are 15-run means. Recurrent accuracy rows are imported summary values whose raw prediction files were unavailable for this revision. MMAC counts operations after the landmark frontend, profiled separately in Table 2. Red: best; bold: second-best, among the image-based models (Image-CNN variants and Ours). EAR rows are landmark-only references excluded from this ranking; the EAR rule’s “~0” denotes a few divisions on six landmark points, not a free method.

Method	PR-AUC $\uparrow$	ROC-AUC $\uparrow$	Acc $\uparrow$	Prec $\uparrow$	Rec $\uparrow$	F1 $\uparrow$	MMAC $\downarrow$	Params $\downarrow$
EAR (rule)	0.8931 $\pm$ 0.0136	0.9169 $\pm$ 0.0146	0.8505	0.8268	0.8878	0.8550	~0	0
EAR + head	0.9724 $\pm$ 0.0096	0.9738 $\pm$ 0.0113	0.9276	0.9312	0.9228	0.9268	0.05	5,009
Image-CNN (max) [2]	0.9114 $\pm$ 0.0291	0.9404 $\pm$ 0.0207	0.8917	0.8526	0.9480	0.8971	31.81	470,561
Image-CNN (+head)	0.9906 $\pm$ 0.0037	0.9930 $\pm$ 0.0036	0.9651	0.9713	0.9584	0.9644	31.85	476,161
Image-CNN (+BiLSTM)	0.9850 $\pm$ 0.0060	0.9889 $\pm$ 0.0048	0.9542	0.9559	0.9524	0.9536	31.89	480,113
Image-CNN (+LSTM)	0.9819 $\pm$ 0.0199	0.9891 $\pm$ 0.0069	0.9615	0.9571	0.9665	0.9615	31.85	475,889
Ours ($D = 16$)	0.9886 $\pm$ 0.0038	0.9914 $\pm$ 0.0037	0.9620	0.9605	0.9636	0.9618	12.49	84,049

temporal head reaches 0.9906 ± 0.0037 against our 0.9886 ± 0.0038 . Our claim is about cost, not accuracy. The paired subject-clustered difference between the two is -0.0040 with a 95% CI of $[-0.0063, -0.0020]$. That interval excludes zero, so the proposed model is measurably the less accurate of the two, but lies inside the prespecified margin $\delta = 0.02$; on the same folds it uses $5.7\times$ fewer parameters (84,049 against 476,161) and $2.6\times$ fewer operations per frame.

The mean of the 15 paired run differences is -0.0019 ± 0.0024 . Pooling scores from different fitted models and averaging their AP values are different estimands; their values must not be interchanged. In a post-hoc margin sensitivity analysis, the primary pooled interval also supports $\delta = 0.01$, but not $\delta = 0.005$. The original 0.02 margin was informed by reference variability, not independently validated as a clinically or operationally acceptable loss.

A second post-hoc analysis gives each subject equal weight: average that subject’s AP over its three source runs, then average the 57 subjects. The paired difference is -0.0062 (subject bootstrap 95% CI $[-0.0114, -0.0023]$). This supports 0.02 but not 0.01, showing that the stronger margin conclusion depends on weighting. These intervals condition on the saved fits and do not capture full retraining uncertainty.

The recurrent summary rows suggest sensitivity to head choice, but do not support new paired inference without their raw predictions. A confidence interval excluding zero cannot be dismissed merely because its effect is smaller than a seed standard deviation. The Image-CNN backbone accounts for essentially all of the recurrent variants’ static computational cost. Its 6,912-to-64 dense layer alone has 442,432 parameters;

the reported parameter ratio is therefore specific to this adapted reference and does not establish superiority over all lightweight CNNs.

The exploratory *v-drop* ablation described under *Encoder Design and Vertical-Resolution Ablation* gives a paired PR-AUC estimate of $+0.0010$ for the proposed encoder (95% CI $[-0.0032, +0.0069]$) across folds 0–1 and three seeds (21 subjects); covering only two of five folds, this is descriptive rather than evidence that vertical resolution is irrelevant, and we retain the proposed encoder.

Robustness across Conditions

Against EAR + head, the pooled-prediction advantage is $+0.0151$ overall. The exploratory analyses include acquisition batch (two groups), glasses (two groups), and brightness, contrast and sharpness terciles (nine groups). Their unadjusted intervals describe observed heterogeneity; they are not simultaneous guarantees of robustness. Against Image-CNN (+head), the available paired subgroup analysis covers acquisition batch and glasses, plus the overall result. Those intervals meet $\delta = 0.02$; corresponding photometric comparisons were not available. Natural subgroup differences do not replace controlled tests of illumination, pose or viewing distance.

Model Complexity and Edge Performance

Frame read (~ 1 ms) and landmark detection (8.3–8.5 ms) are shared by all modes and account for most of the crop-to-e2e gap; EAR (rule), having no encoder, is excluded from the *crop* comparison. The Image-CNN graphs were exported with random weights for structural latency and power characterization, so these

TABLE 2. Per-Frame Latency and Power/Energy Cost on a Raspberry Pi 5, measured together over three free-running repeats (98.65–130.39 s each). Latencies are ms/frame. Crop = eye-crop extraction; Enc. = encoder forward pass; Dec. = decision stage; Sub. = sum of stage medians, not a measured subtotal percentile; e2e99 = 99th-percentile end-to-end latency; $\Delta P(W)$ = active-minus-idle rail power; mJ/frame = total-active rail energy. Rail definition is unverified. Lower is better; bold marks the best learned model in each column.

Method	Crop	Enc.	Dec.	Sub.
EAR (rule, ref.)	0.08	—	0.14	0.22
Ours	0.67	0.86	0.25	1.78
Image-CNN (max)	0.67	2.04	0.17	2.88
Image-CNN (+head)	0.67	2.05	0.31	3.03
Image-CNN (+BiLSTM)	0.67	2.05	0.36	3.08
Image-CNN (+LSTM)	0.67	2.04	0.33	3.03
Method	e2e99	$\Delta P(W)$	mJ/frame	
EAR (rule, ref.)	10.21	2.155	40.70	
Ours	11.88	2.000	46.15	
Image-CNN (max)	12.96	2.260	53.77	
Image-CNN (+head)	13.17	2.209	53.88	
Image-CNN (+BiLSTM)	13.24	2.200	54.38	
Image-CNN (+LSTM)	13.17	2.202	54.36	

measurements do not demonstrate the runtime/energy of the trained accuracy checkpoints in Table 1. At the representation stage the proposed encoder costs 0.86 ms per frame against 2.05 ms for the Image-CNN (+head) control, a 2.4 $\times$ reduction, close to the 2.6 $\times$ static ratio in multiply–accumulate operations; over the method-specific stages together (crop, encode, decide) the ratio is 1.7 $\times$.

End to end, the proposed system has a median latency of 11.27 ms and the control 12.51 ms, i.e. the control is 11% slower. The shared landmark stage accounts for 66–75% of these medians. The observed 99th percentiles are below the 33.3 ms frame period, supporting processing headroom on this replay. This does not establish live-camera deadline reliability or a bound on onset-to-alert delay.

Under free-running replay, the proposed model draws 2.000 W above the measured idle rail sum and uses 46.15 mJ/frame of total active rail energy, versus 53.77–54.38 mJ/frame for the image-CNN variants. Different achieved frame rates, short runs, unverified rails and random-weight comparison graphs prevent a claim of energy savings at a matched 30 fps workload or an inferred battery lifetime.

The latency decomposition is the more direct

TABLE 3. Descriptive external pilot: one EyeBlink8 video. Means over 15 source fits; these fits are not 15 independent external subjects. AP refers to sampled windows; R, F1 and false events/minute (FA/min) refer to continuous detection.

Method	AP	R	F1	FA/min
Ours	0.999	0.998	0.889	3.81
Image-CNN (+head)	0.999	0.992	0.867	4.93
EAR + head	0.965	1.000	0.691	13.28
EAR (rule)	0.998	1.000	0.875	4.21

system result. From the earlier mean latencies, even eliminating the 0.868 ms encoder would reduce the 11.300 ms pipeline by only 7.7%, assuming all other stages stay fixed. Halving the 8.487 ms landmark stage would instead save 37.6% under that same assumption. These are arithmetic upper-bound scenarios, not measured frontend optimizations.

External Continuous-Video Pilot

We additionally evaluate the locally available original EyeBlink8 video 9 [11], [12]: one person, 41 annotated blink IDs. This is one of eight benchmark videos and had previously been used without accuracy labels for timing. The container and timestamp file list 5,183 frames, but 5,134 decode, and the annotation ends at decoded frame 5,133 (zero-based). We discard the 49 unmatched tail timestamps. Nearest-frame resampling using capture times gives 5,133 positions at 30 Hz, with 174 duplicates and 175 unused native frames. MediaPipe crop coverage is 99.75%; all annotated evaluation windows meet the 14-of-19 validity rule. Sampled annotation/eye-corner alignment and checkpoint reproduction were checked as implementation audits.

All 15 source-trained checkpoints and their source-validation thresholds are frozen. No external training, threshold selection or best-run selection is performed. The window diagnostic uses 41 blink-centered windows and 210 deterministic, non-overlapping negative windows without blink overlap. Continuous evaluation thresholds every trailing window and groups consecutive positives, without tuned hysteresis. Predicted intervals are aligned to the window center and matched one-to-one to overlapping blink-ID intervals. All 41 blinks are included over 2.80 annotated minutes. This bilateral-ID protocol is distinct from the dataset authors’ per-eye completeness evaluation.

The proposed model’s continuous F1 varies from 0.726 to 0.976 across source fits (mean 0.889, SD 0.063), despite a mean window AP of 0.999. It pro-

duces 3.81 unmatched detections/minute on average (SD 2.57), and its negative-frame positive rate is 0.63%. Requiring temporal intersection-over-union of at least 0.1 gives the same proposed-model F1 here. The mean of run-wise median onset-to-first-decision offsets is 0.307 s, computed from capture timestamps and excluding processing, display and queue delays. This pilot demonstrates a measurable classification-to-continuous-detection gap; its single subject cannot establish cross-dataset superiority or a population confidence interval.

Discussion and Limitations

The failure on held-out subject U1 remains unresolved: its mean AP is 0.8200 versus 0.9433 for Image-CNN (+head) and 0.9802 for EAR + head. Existing diagnostic checks do not establish a causal explanation or exclude all annotation/preprocessing effects. U1 is retained in every primary analysis; removing it changes the pooled image-control difference only from -0.00402 to -0.00388 , so it does not explain the full trade-off.

The primary benchmark classifies fixed 19-frame events, not continuous detections. Its balanced negative sampling and validation accuracy threshold do not determine the false-alarm rate at deployment prevalence. The nominal 0.3 s center offset is separate from compute latency and from measured alert delay. The primary width was fixed at $D = 16$ before final comparisons; exploratory $D = 8, 32, 64$ results on two folds exist, so the width was not wholly unswept. They do not establish a global optimum. The vertical-stride ablation likewise does not support the proposed geometry as necessary. The one-subject external pilot does not establish generalization beyond the 57-subject source subset; landmark-source shift is not resolved by using the same crop code.

CONCLUSION

A compact encoder with a shared temporal head yields a measured accuracy–cost trade-off on subject-disjoint mEBAL2 events: the pooled AP loss is small but nonzero, and falls within the prespecified 0.02 margin at $5.7\times$ fewer parameters and $2.6\times$ fewer operations than the adapted Image-CNN control. Both frontends use cached 16-D features. Raspberry Pi 5 replay shows ample compute headroom, while its dominant landmark stage limits the benefit of encoder compression. Stronger deployment claims require continuous external evaluation and matched-rate, verified

power measurements.

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